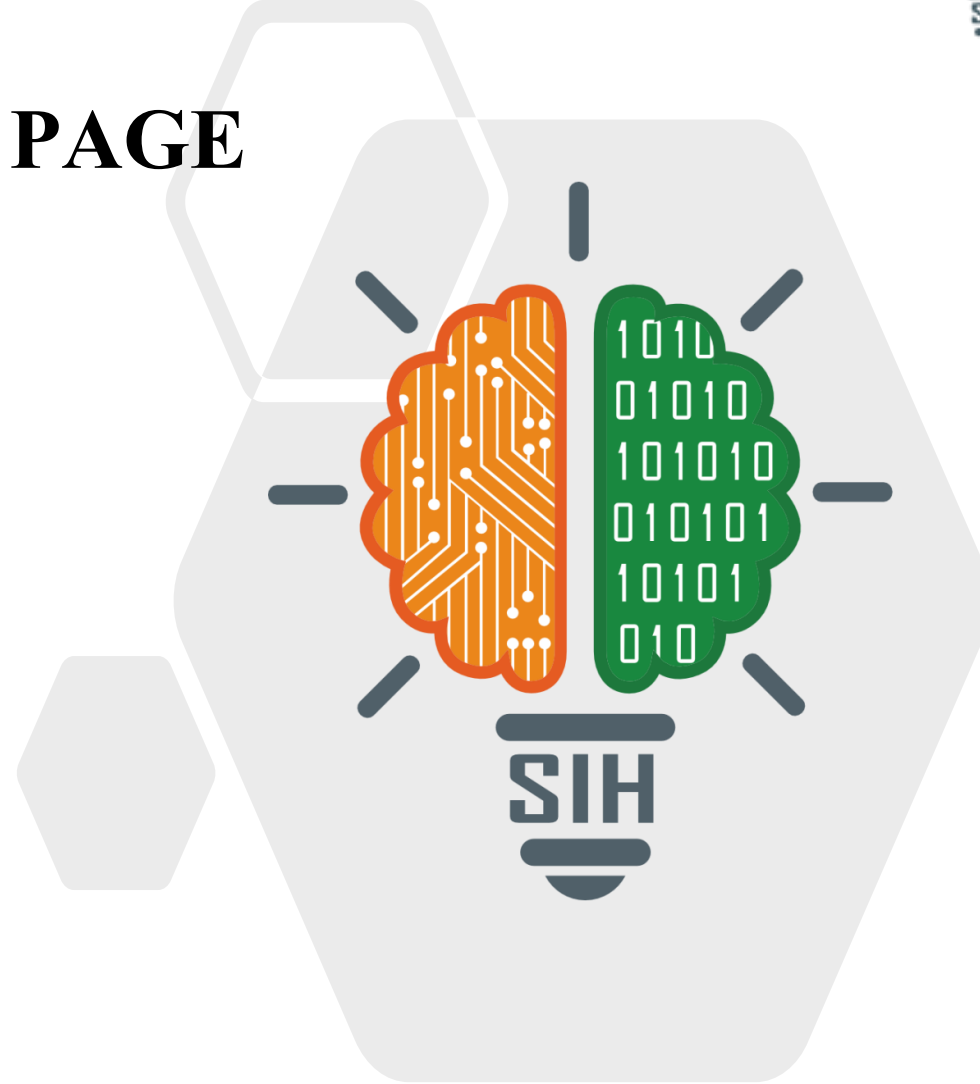


SMART INDIA HACKATHON 2025



TITLE PAGE

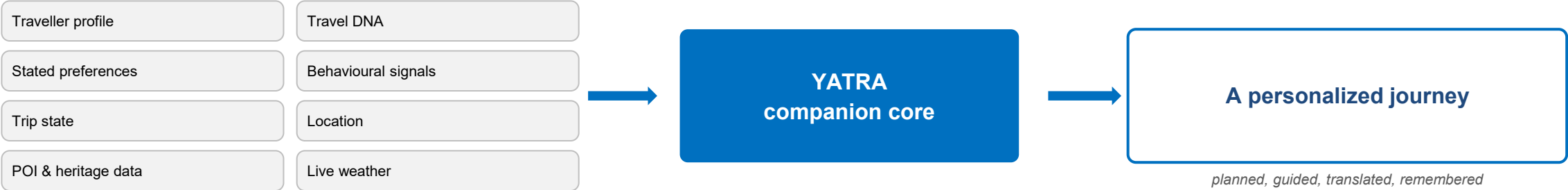
- Problem Statement ID –
- Problem Statement Title-
- Theme-
- PS Category- Software/Hardware
- Team ID-
- Team Name (Registered on portal)



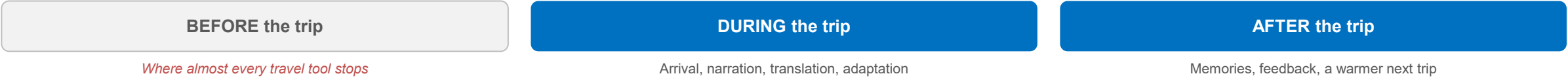
ChatGPT can plan a trip. YATRA travels it with you.

An AI travel companion for the whole journey — not an itinerary generator that stops at the plan.

WHAT YATRA COMBINES INTO ONE COMPANION



THE GAP WE CLOSE



WHY A GENERAL CHATBOT IS NOT THIS

the difference is the product system around the model, not the model

Invented places are structurally impossible	Confidence is measured, not self-reported	The validator overrules the model	Degradation is a designed feature
Candidates are numbered before the prompt is built. An index outside that range is rejected before anything is saved — proven by a live test asserting every scheduled place exists in the database set.	It is derived from retrieval similarity, then shown to the traveller as “Not fully verified — please confirm locally.” We never ask the model whether it is sure.	An AI-proposed change that creates a new overlap or an impossible travel time is vetoed, and the conflict is explained back to the traveller instead.	Every AI path has a real non-AI path, and the app is told when it is on one. The product still works on the day the model does not.

TECHNOLOGIES · ONE STACK, TOP TO BOTTOM

MOBILE

React Native + Expo · TypeScript

28 screens · 6 UI languages · session in device secure storage

API

FastAPI /v1

21 routers · 73 endpoints · ES256 token verification via JWKS · one response envelope · per-user rate limit

SERVICES

35 modules

itinerary · modification · narration · photo Q&A · personalization · business rules · location · safety · budget · group

LLM GATEWAY

one Protocol: `complete` · `complete_multimodal` · `complete_audio` · `embed`

Gemini adapter

the only file that imports a vendor SDK

second provider

one adapter class — the vendor is a config value

DATA

Supabase Postgres

38 tables · Row-Level Security on 37 · PostGIS · pgvector(1536)

EXTERNAL

Geoapify Places

Open-Meteo

Supabase Storage

Expo Push

ENGINEERING GUARANTEES

- A vendor SDK is imported in exactly one file
- 9 versioned prompt modules — no prompt is written inline in a service
- Every structured output passes through one parse-and-validate choke point
- Confidence comes from retrieval similarity — never from asking the model if it is sure

METHODOLOGY · HOW A PLAN IS BUILT

- 1 Rule-filtered candidates pulled from the database
- 2 The prompt carries them as a numbered list
- 3 Structured output, validated against a schema
- 4 Any out-of-range index is rejected
- 5 Deterministic rule validator runs
- 6 Persisted, day by day

The model composes. The database and the validator decide.

WHEN THE AI FAILS

Model responds

Model fails or times out

Deterministic plan from real database places

Honest error

It never fabricates a success.

FEASIBILITY · ALREADY BUILT AND RUNNING

we mark our own gaps rather than hide them

● Live

● Live, limited

● Built, awaiting a credential

○ Roadmap

IDENTITY & PROFILE	PLANNING & AI	ON-SITE COMPANION	HERITAGE & LANGUAGE	TRIPS, MEMORY & SOCIAL
● Google sign-in · JWKS verified ● Onboarding → traveller profile ● Travel DNA ● Row-Level Security, 37 tables	● AI itinerary generation ● Idea extraction from free text ● Conversational modification ● Business-rule validator ● Budget estimate & expense split ● Group planning & reconciliation ● Quick Plan (local outing) ○ Learned ranking	● Location consent & arrival ● Nearby recommendations ● Live weather ● Maps & POI discovery ● Weather-driven adaptation ● Safety / SOS & location share ○ Background tracking ○ Proactive companion	● Grounded narration + citation ● Confidence flagging ● Photo Q&A (multimodal) ● Translation, 5 languages ● Speech translation (batch) ● Curated phrasebook ○ Semantic section navigation	● Trips, favourites, collections ● Reviews + admin moderation ● Memory Box + private storage ● Notifications ● Offline package ○ Booking & payments ○ AI trip recap

EVIDENCE

73 versioned API endpoints across 21 routers, one error envelope	38 tables · 22 migrations · PostGIS + pgvector
560 automated tests (187 mobile + 373 backend)	367 passed against live Postgres and a real model key, 06 Sep 2026

That live run also exposed four real defects in our own code. We fixed them and kept the record.

CHALLENGES → HOW WE ALREADY HANDLE THEM

Model cost or quota runs out	Provider abstraction · per-user rate limit · a deterministic non-AI plan built from real database places
The model invents a place or a fact	Candidates are numbered; an out-of-range index is rejected before it is saved · narration is grounded in curated content
A third-party service goes down	Every provider client collapses to one typed error · the client is told it is degraded, never shown a fake success
The content catalogue is thin	Curated-first seed scripts · admin publish flow · live Places augmentation when the cache is sparse

VIABILITY	Hosted models only, nothing trained · fast-tier model, caching and rate limits · booking-partner path exists in the schema but is not required to prove the MVP	11 curated heritage sites across 9 cities today Target for launch: 6–7 destinations, 40–50 points of interest
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WHAT CHANGES FOR THE TRAVELLER

TODAY

Search

Maps

OTA

Reviews

Notes

₹ Guide

plan goes stale

YATRA



BENEFITS

For the traveller (social)

- Grounded narration on demand, with its source named — instead of a per-monument guide fee
- Phrase help in 5 Indian languages with transliteration
- Arrival awareness, and a manual “I’m here” when GPS won’t cooperate
- One conversation instead of five open tabs

For destinations (economic)

- Places are matched to one traveller’s stated interests from a curated catalogue — not ranked by what is trending this week
- The same pipeline that surfaces the Taj can surface Hampi or Ajanta
- A mechanism for wider discovery — not a promise to rescue heritage

For the product (sustained use)

- A companion has a reason to be opened during the trip, not only before it
- Every accepted, skipped or corrected suggestion is captured as a signal
- The next trip starts less cold than the last one

ONE TRAVELLER, END TO END *Meera · 26 · solo · Delhi + Agra · Mughal architecture · no Hindi · ~₹15,000*

Signs in

Five onboarding screens become a profile that persists

Pastes her ideas

“The Taj at sunrise” matched to a real place row

Asks for a plan

Model composes; budget and timing checked deterministically

“Make it relaxed”

Two stops rescheduled, the rest left untouched

Lands in Delhi

Per-trip consent, arrival — or a manual “I’m here”

Stands at the Taj

Grounded narration, a photo question, a Hindi phrase

Comes home

Memories kept, feedback recorded, next trip warmer

HOW WE WOULD MEASURE IT *PRD §37 — metrics, not adjectives*

Itinerary generation success rate

Recommendation acceptance rate

Itinerary modification rate

Engagement across trip lifecycle

Narration engagement rate

Hallucination-flag rate

These are mechanisms we built and metrics we will measure — not outcomes we are claiming.

WHAT THE RESEARCH SAYS *each finding traces to a section of our Master PRD & SDLC document*

Planning is fragmented

A search engine, a map, an OTA, review sites and personal notes — no single tool holds the full picture.

PRD §3.3

On-site context is a plaque or a fee

Historical understanding is either two lines on a board or a paid guide. One guided visit to a major monument runs roughly ₹1,000.

PRD §4.3

The language barrier is domestic

In India a traveller doesn't need to leave the country to lose the language — only the state. Translation products are built for cross-border travel.

PRD §3.2

The app goes quiet on arrival

Usage spikes during planning, then the tool goes unopened for the part of the trip that actually matters.

PRD §4.6

WHO WE DESIGNED FOR *Five planning patterns — not five demographics.*

Meera

Solo culture-seeker Delhi + Agra, doesn't speak Hindi

The Weekend Squad

4–5 friends, short notice, shared budget

The Family Traveller

Comfortable pace, low-surprise logistics

The Heritage Enthusiast

Goes deeper than a plaque, asks specific questions

Arjun

Everyday local plans, not only multi-day trips

REFERENCES AND SOURCE MATERIAL

- **Master PRD & SDLC — AI Personalized Tourist Guide (TC-SO1)**
74 pages · §3 Business Context, §4 Problem Statement, §9 Personas, §29 MVP, §37 KPIs, §44 Innovation
- **UNESCO World Heritage Centre — India**
whc.unesco.org/en/statesparties/in — heritage site records used to source narration content
- **Archaeological Survey of India**
asi.nic.in — monument records and site citations carried on every narration section
- **Ministry of Tourism, Government of India**
tourism.gov.in — India Tourism Statistics, domestic travel context
- **Census of India — language tables**
censusindia.gov.in — evidence for the domestic, inter-state language barrier
- **Google Gemini API · Open-Meteo (WMO codes) · Geoapify · Supabase · pgvector · PostGIS**
technical documentation for every external service integrated in the working prototype

We assert no market-size figure. The PRD deliberately leaves it to be confirmed rather than inventing precision — so does this deck.